39 A nucleus X undergoes β^+ decay.

The products are a nucleus Y with proton number Z, a β^+ particle and another particle P.

$$X \rightarrow {}_{Z}Y + \beta^+ + P$$

What is particle P and what is the proton number of nucleus X?

	P	proton number of nucleus X
A	antineutrino	$Z - 1$
B	antineutrino	$Z + 1$
C	neutrino	$Z - 1$
D	neutrino	$Z + 1$